Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/19/2016
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (NNX14AB31G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2013	**End Date:**	06/30/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Bhat, Biliyar	**Contact Phone:**	
Contact Email:	biliyar.n.bhat@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2016 per NSSC information (Ed., 1/15/16) NOTE: End date is now 3/31/2016, per NSSC information (Ed., 9/9/15)		
Key Personnel Changes/Previous PI:	May 2016 report: After the postdoctoral researcher Dr. Jose A. Alvarado-Contreras, involved in computer simulations, returned in fall 2014 to his university position, the code development was completed by Dr. Andrey Maximenko. October 2014 report: Mr. Timothy Young, previously MS student on the project, returned as a part-time engineering researcher to prepare all ground and flight samples. Mr. Hari Kishor took over microstructure and hardness testing, developing techniques appropriate to the small samples. The postdoctoral researcher Dr. Jose A. Alvarado-Contreras, involved in computer simulations, has had to return to his university position and the code development was completed by Dr. Andrey Maximenko.		
COI Name (Institution):	Olevsky, Eugene (San Diego State University)		
Grant/Contract No.:	NNX14AB31G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	NOTE (Ed., January 2014): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments", grant # NNX10AV38G with the same Principal Investigator. A three year research project is proposed to investigate the gravity-dependent densification, microstructure development, and the related macroscopic shape evolution of multi-phase powder materials during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as a technique for in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team has developed a computer code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of tungsten heavy alloys (W-Ni-Cu-Mn) to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is responsible for approximately $25 billion in annual product for applications that include components used in metal cutting, oil well drilling, automotive engines, motor bearings, computer heat sinks, jet engine valves, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. During that high temperature sintering the component strength drops considerably and is subject to gravity effects. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a computational model that predicts final size and shape and with multiple scale model predicts properties. Further, with robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions as required for eventual fabrication on the moon, planets, or in microgravity, for example, in building structure on the lunar surface or repairing space vehicles.
Task Progress:	The last two years resulted in significant progress on documenting microgravity sintering behavior, publishing the current models, fabrication of ground and flight samples (with full characterization), developing evaluation protocols, and documenting the novel tungsten heavy alloys developed for the furnace on the International Space Station. Careful consideration has been given to replication data available for liquid phase sintering on Earth and parallel, repeat experiments in microgravity. Considerable experience exists with sintering on Earth, but the behavior under reduced gravity conditions is poorly understood. Our study analyzes replica microgravity liquid phase sintering data for seven tungsten alloys, sintered for three hold times to 10 h at 1500°C using 0.002% of standard gravity. Equivalent sintering is performed on Earth. Microgravity sintering results in a lower density and more shape distortion. For Earth-based sintering, minimized distortion is associated with low liquid contents to avoid solid settling and slumping. Distortion in microgravity sintering involves viscous spreading of the component at points of contact with the containment crucible. Microgravity sintering exhibits pore coalescence into large, stable voids that cause component swelling. The microgravity sintering results show good replication in terms of mass change and sintered density. Distortion is scattered but statistically similar between the replica microgravity runs. However, subtle factors, not typically of concern on Earth, emerge to influence microgravity sintering. To accommodate the furnace capabilities on the International Space Station, new liquid phase sintering formulations were required. Tungsten heavy alloys are desired because of the large gravity effect on densification and distortion. Efforts to reduce the sintering temperature, especially for room temperature applications, resulted in discovery that transition metal additions were effective via liquid phase sintering and activated sintering. The sintering temperature drops to about 1500°C. This is because the transition metal shortcuts the diffusive transport process required for densification. In that context, new tungsten alloys containing nickel, copper, tin, and manganese further lower the sintering temperature to about 1200°C. Full density is possible using sintering temperatures applicable to steels. Issues in alloy design, densification kinetics, grain coarsening, and distortion are the focus of current research using 1200°C sintering cycles in anticipation of critical microgravity sintering on the International Space Station. The fabrication of flight and ground samples has been documented and finalized. All required samples are fabricated, and trials have been conducted for vapor containment, sample adhesion to the crucibles, toxic vapor release, and other required trials. All tests have succeeded in showing the planned flight experiments are safe and within capabilities. Recent efforts have focused on sample evaluation, including microstructure, density, dimensions, and hardness. This involves various ground samples to set a protocol. Hardness data are in hand and will be included in new publications. EDITOR'S NOTE (May 2016): Project continues with the same title, "MULTI-SCALE MODELING AND EXPERIMENTATION ON LIQUID PHASE SINTERING IN GRAVITY AND MICROGRAVITY ENVIRONMENTS," grant NNX16AK21G, to begin in summer 2016.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Articles in Peer-reviewed Journals	German RM, Johnson JL. "Replication experiments in microgravity liquid phase sintering." Metallurgical and Materials Transactions. 2016 May; 47A(5):2286-99. http://dx.doi.org/10.1007/s11661-016-3372-3 , May-2016

Articles in Peer-reviewed Journals	German RM. "Sintering trajectories: Description of how density, surface area, and grain size change." Journal of Metals. 2016 Mar;68(3):878-84. http://dx.doi.org/10.1007/s11837-015-1795-8 (published as invited article on Special Topics in Sintering) , Mar-2016
Articles in Peer-reviewed Journals	German RM. "Sintering simplified: Surface area, density, and grain size relations." Materials Science Forum. 2016 Jan;835:50-75. http://dx.doi.org/10.4028/www.scientific.net/MSF.835.50 (Published as Invited Paper in Special Topics Issue, Ed. G.S. Upadhyaya) , Jan-2016
Articles in Peer-reviewed Journals	German RM. "Lower sintering temperature tungsten alloys for space research." International Journal of Refractory Metals and Hard Materials. 2015 Nov;53 Pt B:74-9. http://dx.doi.org/10.1016/j.ijrmhm.2015.04.020 , Nov-2015
Articles in Peer-reviewed Journals	Bollina R, Suri P, Park SJ, German RM. "Microstructural evolution of a 93 Wt.% tungsten heavy alloy: A quenching study to understand the evolution of contiguity, connectivity with sintering temperature and time." Journal of Powder Metallurgy and Mining. 2015:4(1):130-4. http://dx.doi.org/10.4172/2168-9806.1000130 , Dec-2015
Papers from Meeting Proceedings	German RM. "The Gravitational Role of Liquid Phase Sintering." Presented at the 2015 PowderTech Conference, 2015 International Conference on Powder Metallurgy & Particulate Materials, San Diego, California, May 17-20, 2015. In: Advances In Powder Metallurgy and Particulate Materials – 2015. Proceedings of the 2015 International Conference on Powder Metallurgy & Particulate Materials, May 17-20, San Diego, California. Princeton, NJ : Metal Powder Industries Federation, 2015. p. 5.90-5.104. , Aug-2015